

ALISTRO, France

WMO: 077750

Lat: 42.2597N

Lon: 9.5414E

Elev: 74

StdP: 100.44

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.4	4.9	-7.8	2.0	7.9	-6.0	2.3	8.3	14.5	12.4	12.4	10.0	6.1	340	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	31.2	23.0	30.1	22.9	29.1	22.5	25.0	29.4	24.3	28.6	23.7	27.8	4.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.5	28.2	22.9	17.8	27.6	22.2	17.0	26.9	76.3	29.5	73.7	28.7	71.1	27.9	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	9.8	8.3	DB	1.8	33.1	1.7	1.9	0.5	34.4	-0.4	35.5	-1.4	36.6	-2.6	37.9	(4)
(5)			WB	-0.8	25.6	1.2	0.8	-1.7	26.2	-2.4	26.7	-3.1	27.1	-4.0	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	10.8	10.5	12.3	15.0	18.1	22.5	25.3	25.5	22.3	18.9	14.8	12.0	(6)	
	DBStd	5.81	2.50	2.70	2.36	2.04	2.17	2.53	1.87	1.89	2.17	2.23	2.59	2.56	(7)	
	HDD10.0	69	21	24	9	1	0	0	0	0	0	0	2	12	(8)	
	HDD18.3	1101	235	221	186	102	29	2	0	0	2	19	108	197	(9)	
	CDD10.0	2761	44	37	81	150	252	376	473	481	370	276	146	74	(10)	
	CDD18.3	751	0	0	0	2	23	127	215	223	122	37	2	0	(11)	
	CDH23.3	4727	0	0	0	2	58	695	1720	1751	468	34	1	0	(12)	
	CDH26.7	1110	0	0	0	0	3	138	437	472	60	1	0	0	(13)	
(14) Wind	WSAvg	3.9	4.2	4.3	4.4	3.9	3.8	3.5	3.4	3.4	3.7	3.8	4.1	4.1	(14)	
(15) Precipitation	PrecAvg	705	65	53	65	55	48	28	12	17	66	107	105	83	(15)	
	PrecMax	1170	181	148	218	111	201	90	63	60	193	270	270	174	(16)	
	PrecMin	560	8	0	3	8	9	0	0	0	4	3	7	1	(17)	
	PrecStd	155	42	42	58	28	43	22	17	15	56	70	65	53	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	18.3	20.3	22.7	26.9	31.9	32.5	33.7	30.3	25.6	22.0	19.8	(19)	
		MCWB	11.8	12.0	12.6	15.5	18.1	21.5	23.1	23.2	20.7	19.6	16.8	13.4	(20)	
	2%	DB	16.6	16.2	18.2	21.2	24.9	29.5	30.9	31.5	28.2	24.3	20.3	17.6	(21)	
		MCWB	11.4	11.3	12.2	15.2	18.5	21.8	23.4	23.2	21.7	19.3	16.1	12.7	(22)	
	5%	DB	15.1	15.1	16.9	20.0	23.4	28.2	29.9	30.4	27.1	23.2	19.1	16.4	(23)	
		MCWB	11.0	10.9	11.9	14.3	17.5	21.6	23.1	23.2	21.2	18.9	15.6	12.4	(24)	
	10%	DB	14.1	14.1	15.8	18.7	22.1	27.0	29.0	29.1	26.0	22.2	18.1	15.3	(25)	
		MCWB	10.4	10.4	11.6	13.8	16.8	21.0	22.6	22.8	20.3	18.4	14.9	11.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.7	14.7	17.3	21.1	24.2	25.8	25.7	24.3	21.9	18.5	15.2	(27)	
		MCDB	16.2	16.1	17.8	21.2	24.3	29.0	30.4	30.5	27.4	23.6	20.2	17.2	(28)	
	2%	WB	12.7	12.6	13.7	16.2	19.5	23.3	25.0	25.1	23.2	20.8	17.4	14.4	(29)	
		MCDB	15.1	14.8	16.4	19.6	22.8	27.6	29.5	29.6	26.6	22.7	19.2	16.4	(30)	
	5%	WB	11.9	11.9	13.1	15.4	18.5	22.5	24.3	24.5	22.3	20.0	16.5	13.4	(31)	
		MCDB	14.3	14.2	15.4	18.5	22.0	26.7	28.7	28.8	25.9	22.3	18.2	15.6	(32)	
	10%	WB	11.1	11.0	12.5	14.7	17.7	21.7	23.7	23.8	21.4	19.2	15.7	12.5	(33)	
		MCDB	13.5	13.3	14.7	17.6	21.3	25.9	27.9	28.0	25.0	21.5	17.5	14.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.7	5.9	6.2	6.5	6.8	6.7	6.6	6.2	5.6	5.3	5.3	(35)	
		MCDBR	6.6	6.9	7.6	7.7	7.8	7.7	7.4	7.7	7.2	6.6	6.0	6.3	(36)	
	5% WB	MCWBR	4.9	5.3	5.8	5.2	5.4	5.2	5.3	5.9	5.3	4.9	4.5	4.8	(37)	
		MCWBR	5.9	6.3	6.5	6.7	6.7	6.9	6.9	7.0	6.6	5.7	5.1	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.323	0.340	0.369	0.405	0.416	0.408	0.407	0.404	0.395	0.376	0.354	0.326	(40)	
		taud	2.524	2.467	2.401	2.304	2.316	2.378	2.382	2.400	2.418	2.466	2.473	2.522	(41)	
	Ebn at Noon		824	863	875	864	861	867	863	855	837	809	773	786	(42)	
		Edh at Noon	72	88	105	125	127	119	118	112	103	88	75	67	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.73	2.63	3.84	5.14	6.20	7.02	7.16	6.23	4.69	3.14	1.94	1.54	(44)	
	RadStd		0.16	0.25	0.40	0.31	0.46	0.35	0.29	0.30	0.22	0.27	0.17	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+71	N/A	(47)	

Nomenclature: See separate page